

SIMATS ENGINEERING

ASSESSMENT TOOL 3 : Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Jani Dilip
Register Number	192424314

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

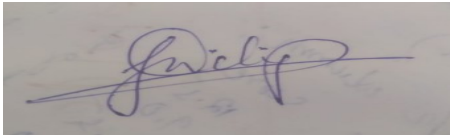
Total Marks: 50

Code of Conduct

I **Jani Dilip (192424314)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

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Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define Cloud Computing and Explain Any Four Characteristics

Cloud computing is the on-demand delivery of computing services such as servers, storage, databases, networking, software, and analytics over the Internet. Users can access these services without owning or managing physical infrastructure and pay only for what they use.

Characteristics

On-Demand Self-Service

Users can automatically provision computing resources whenever required without interacting with the service provider.

Example: Launching a virtual machine in Microsoft Azure within minutes.

Broad Network Access

Cloud services are available over the Internet and can be accessed using laptops, smartphones, tablets, or desktops from anywhere.

Example: Accessing Google Drive from multiple devices.

Resource Pooling

Cloud providers share computing resources among multiple users (multi-tenancy), allocating resources dynamically according to demand.

Example: Amazon Web Services (AWS) serves thousands of customers using shared infrastructure.

Rapid Elasticity (Scalability)

Resources can be increased or decreased quickly depending on workload requirements.

Example: An online shopping website automatically adds servers during a festival sale.

2. Evolution of Cloud Computing

Cloud computing evolved through several technological stages.

Hardware Evolution

Large mainframe computers were used in the 1950s–1960s.

Computing resources were expensive and centralized.

Personal Computing Era

Personal computers became affordable.

Organizations started maintaining their own servers.

Networking Evolution

Local Area Networks (LAN) and Wide Area Networks (WAN) enabled computers to communicate and share resources.

Internet Evolution

The Internet connected millions of systems worldwide.

Web-based applications became popular.

Virtualization Technology

Physical servers were divided into multiple virtual machines.

Resource utilization increased significantly.

Internet Software Evolution

Software shifted from installation on local computers to browser-based applications.

Services such as Gmail and Microsoft Office 365 became available online.

Result

These developments enabled modern cloud platforms such as:

Amazon Web Services (AWS)

Microsoft Azure

Google Cloud Platform (GCP)

They provide scalable, reliable, and on-demand computing services.

3. Role of Server Virtualization in Cloud Computing

Server Virtualization

Server virtualization is a technology that allows a single physical server to be divided into multiple Virtual Machines (VMs) using software called a hypervisor. Each virtual machine operates independently with its own operating system and applications. Virtualization improves hardware utilization, reduces costs, and increases flexibility, making it a key technology for cloud computing.

Role of Server Virtualization in Cloud Computing

Improves Resource Utilization: Multiple virtual machines share the same physical server, reducing hardware wastage.

Reduces Cost: Fewer physical servers are required, lowering hardware, power, and maintenance costs.

Supports Scalability: Virtual machines can be created, resized, or removed quickly based

on user demand.

Enables Rapid Provisioning: New servers can be deployed within minutes instead of purchasing new hardware.

Provides Isolation: Each virtual machine runs independently, so problems in one VM do not affect others.

Enhances Disaster Recovery: Virtual machines can be backed up, migrated, and restored easily.

Forms the Foundation of Cloud Computing: Cloud providers use virtualization to deliver computing resources efficiently.

Example

One physical server can host several virtual machines running Windows and Linux simultaneously.

4. Compare SOAP-Based and REST-Based Web Services

SOAP (Simple Object Access Protocol) and REST (Representational State Transfer) are two popular approaches used to develop web services. Both enable communication between applications over the Internet, but they differ in communication style, data exchange, and suitability for cloud applications.

Communication Style

SOAP is a protocol that follows strict standards and predefined rules for communication.

REST is an architectural style that uses standard HTTP methods such as GET, POST, PUT, and DELETE to access web resources.

Data Exchange

SOAP exchanges data only in XML format, making messages larger and slower to process.

REST supports multiple formats such as JSON, XML, HTML, and plain text. JSON is the most commonly used because it is lightweight and faster.

Suitability for Cloud Applications

SOAP is suitable for applications requiring high security, reliability, and transaction support, such as banking and financial systems.

REST is widely used in cloud, web, and mobile applications because it is simple, scalable, and provides better performance.

Key Differences

SOAP is a protocol, whereas REST is an architectural style.

SOAP uses only XML, while REST supports JSON, XML, HTML, and text.

SOAP provides WS-Security, while REST mainly uses HTTPS and OAuth.

SOAP is slower due to XML processing, whereas REST is faster and lightweight.

Example: SOAP – Banking systems; REST – Google Maps API.

5. Differentiate IaaS, PaaS, SaaS, and XaaS

Cloud computing offers different service models based on the level of control and responsibility required by users. The main service models are IaaS, PaaS, SaaS, and XaaS.

1. Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources such as servers, storage, and networking over the Internet. Users manage the operating system, applications, and data, while the cloud provider manages the hardware.

Key Points:

Provides virtual infrastructure.

Highly scalable and flexible.

Pay-as-you-use pricing model.

Example: Amazon EC2.

2. Platform as a Service (PaaS)

PaaS provides a complete platform for developing, testing, and deploying applications. Developers can focus on coding without managing the underlying infrastructure.

Key Points:

Provides development tools and runtime environment.

No need to manage servers or operating systems.

Faster application development.

Example: Google App Engine.

3. Software as a Service (SaaS)

SaaS provides ready-to-use software applications over the Internet. Users can access the software through a web browser without installing or maintaining it.

Key Points:

Ready-to-use software.

Accessible from anywhere with an Internet connection.

Automatic updates and maintenance.

Example: Microsoft 365.

4. Anything as a Service (XaaS)

XaaS is a broad term that refers to any IT service delivered through the cloud, including storage, databases, AI, networking, and security services.

Key Points:

Includes all cloud-based services.

Flexible and scalable.

Reduces infrastructure costs.

Example: AWS (Amazon Web Services).